

Convex optimization — Exercise sheet I

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Exercise 1

In this exercise, the objective is to model optimization problems, i.e. to write for each of these problems the corresponding formalized optimization program. *Here, there is no need to solve these programs.*

1. Find the length of the base b and the height h of the triangle with the largest possible area and whose perimeter is at most 50 cm. Let us admit here that the perimeter of a triangle is at least $b + \sqrt{4h^2 + b^2}$.

Geometric note (to find the admitted formula): let us consider a triangle ABC where the AB segment, of length b , is the base and h is the height. Let us draw the line L parallel to AB and passing through C. To find the minimum perimeter, since the height and base of the triangle is fixed, we look for the position of C on L which minimizes the distance $AC + CB$. We construct the point D symmetric to B with respect to L ; by construction $CB = CD$, and so $AC + CB = AC + CD$. This distance is minimal when C is placed on the straight line between A and D (triangle inequality), that is when $AC = CD = CB$. Hence the perimeter is minimized when the triangle is isosceles.

When the triangle is isosceles, and according to Pythagorean theorem: $AC = CB = \sqrt{h^2 + \left(\frac{b}{2}\right)^2}$, and thus the perimeter is equal to $b + 2\sqrt{h^2 + \frac{b^2}{4}} = b + \sqrt{4h^2 + b^2}$.

2. A candy manufacturer produces two types of candies. Type I candies are sold for 6 € each, and type II candies for 7 € each. For their manufacture, each unit requires different ingredients detailed in the table below.

	Chocolate	Sugar	Caramel
I	2	3	1
II	0	2	4

Table 1: Units necessary for the manufacture of each type of candies

Moreover, the warehouses of this manufacturer contain 400 units of chocolate, 900 units of sugar and 700 units of caramel. Find out what it has to produce in order to maximize its revenue, assuming that all of its production will be sold.

Exercise 2

For each of the following functions, indicate and justify whether it is convex, concave, both, or neither.

$$f_1(x) = 7x, \quad x \in \mathbb{R}$$

$$f_4(x) = |x|, \quad x \in \mathbb{R}$$

$$f_2(x) = \sin(x), \quad x \in \mathbb{R}$$

$$f_5(x) = x \log\left(\frac{1}{x}\right), \quad x \in \mathbb{R}_{>0}$$

$$f_3(x, y) = 2y - 3x, \quad (x, y) \in \mathbb{R}^2$$

$$f_6(x, y) = \frac{x^2}{2} + \frac{y^2}{2}, \quad (x, y) \in \mathbb{R}^2$$

Exercise 3

1. Prove that a half-space $\{x \in \mathbb{R}^n : a^\top x \leq b\}$ (with $a \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ and $b \in \mathbb{R}$) is a convex set.
2. Prove that the intersection of convex sets is a convex set.

Exercise 4

Try to solve empirically the transportation problem presented in slides 14 to 18 (section 1.1) of the introduction.